Establishing Ground Truth: A Hybrid 3D-Printed & Synthetic Data Generation Framework For Molecular Imaging Assessment

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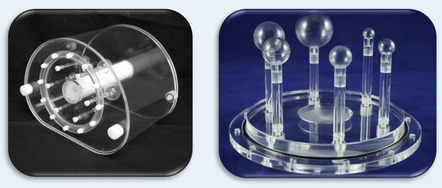
BACKGROUND

Molecular Imaging advances — Dynamic imaging (4D), Theranostics, and AI are evolving rapidly

Need for ground truth — to validate new features, tools, protocols ***prior to*** clinical use

Standardized Phantoms (physical test objects) lack customizability and require tedious preparation highly prone to errors

Limited validation diversity — small clinical datasets and few controlled ground truths restrict robust assessment and confidence of new imaging tools



Typically used quality control (QC) phantoms

Solution: A Complementary Framework

Tool A (Physical): 3D Printed Phantoms to provide physical ground truth

Tool B (Synthetic): Simulation to provide analytical ground truth and extend assessment

METHODS

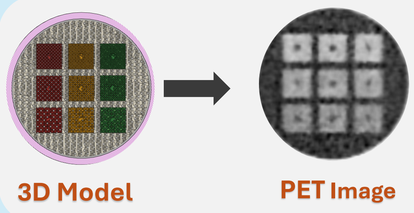
3D Printed "Analog" Ground Truth

The Engineering Constraint: Creating variable image contrast *without* mixing multiple liquid solutions / multiple compartments



Porosity (ϕ) as parameter to control dilution of radiotracer -> adjust image contrast

$$\text{Tissue Activity Concentration } (AC_{Bkg}) = \phi * AC_{Lesion}$$



Lesions appear more darker and tissue brighter

- **Advantages of this Method:** Simple, Customizable, Reproducible, Multi-Contrast with single injection

Diffusion Decay Simulations with Synthetic Noise Addition

The Computational Constraint: Generating scalable, realistic datasets *without* the processing time of full Monte Carlo physics.

Tracer Distribution

$$\frac{\partial C_i}{\partial t} = \nabla.(\mathbf{D}.\nabla C_i) - (\lambda * C_i);\ \lambda = \frac{log\,2}{T_{1/2}}$$

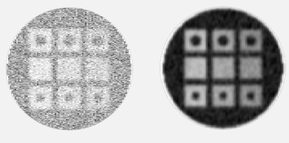
D: Diffusion Coefficient of Glucose
C: Concentration in mol/m3
$T_{1/2}$: Radiotracer Half life



Ideal Map

+

Scanner Noise



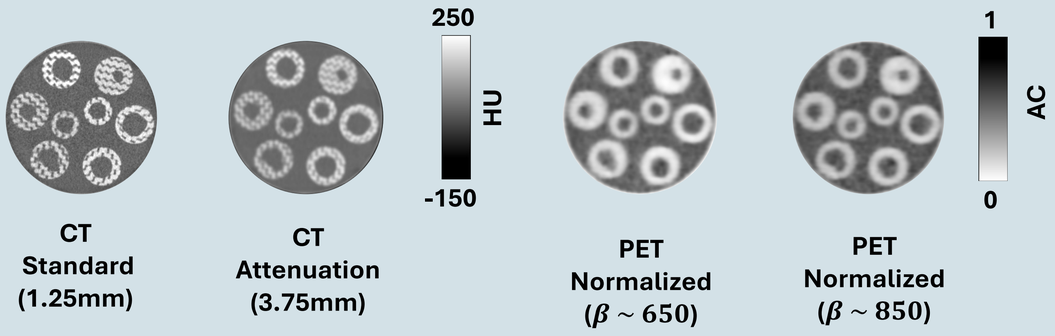
Pseudo Poisson | Post Gaussian

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Simulated Image



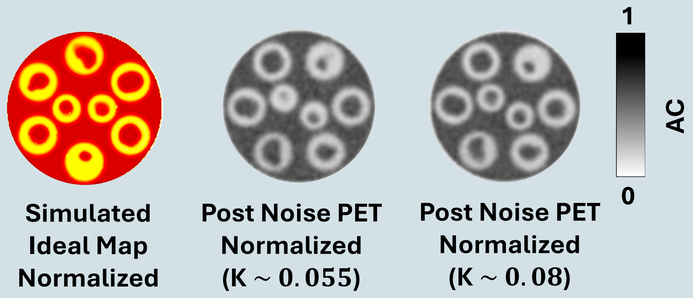
RESULTS



CT Standard (1.25mm) | CT Attenuation (3.75mm) | PET Normalized ($\beta \sim 650$) | PET Normalized ($\beta \sim 850$)

Increase in beta (reconstruction smoothness) suppresses noise at low activity

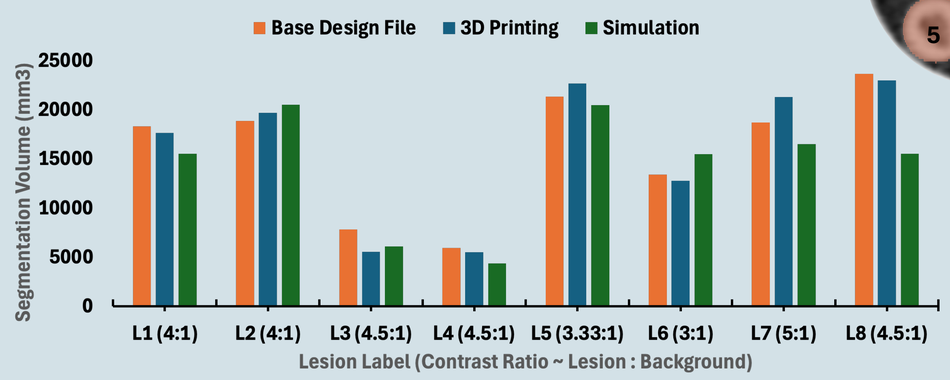
! K is the noise calibration parameter used to match the mean variance ratio of simulation to acquisition



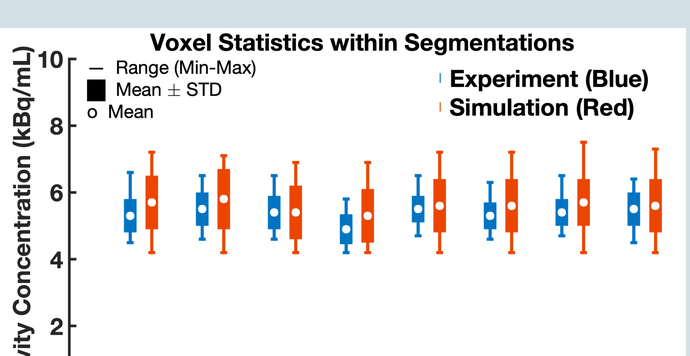
Simulated Ideal Map Normalized | Post Noise PET Normalized ($K \sim 0.055$) | Post Noise PET Normalized ($K \sim 0.08$)

Mean segmentation volume differences were 9% for Experiment and 16% for simulation data – Lower than typically used clinical margin of 20%





For $\beta \sim 850$ & $K \sim 0.08$



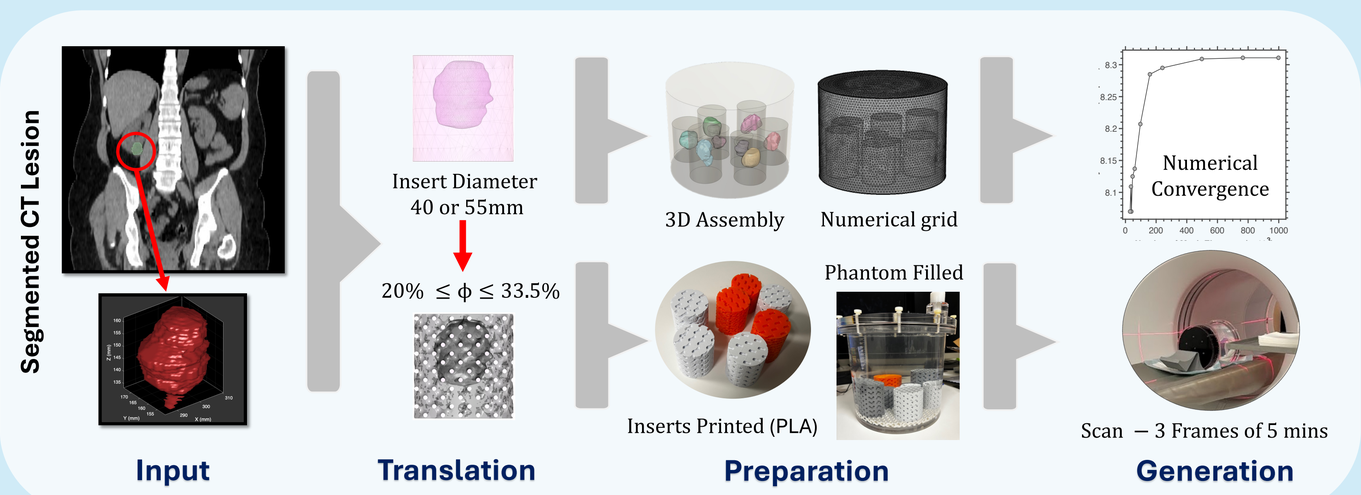
- **Mean simulated values less than 10% variability**
- **Maximum values have 12.5% variability indicating slight overestimation**

The synthetic model provides a reliable "Absolute Truth" for quantitative benchmarking

Anthropomorphic Lesion Evaluation – Pilot Study

8 Patient Lesions extracted from The Cancer Archives (TCIA)

Purpose: Bridging the gap between simple spheres and clinically realistic geometries



50MBq FDG ^{18}F - 45 Minutes prior to scan

Clinical Acquisition & Reconstruction

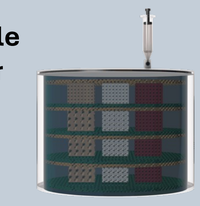
Metrics for Quantitative Inference

65% Isolevel Segmentation
Calibrated to largest pososity (33.5%) with a 2.5% margin

Segmentation Volumes & Statistics

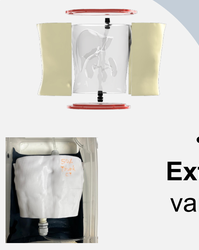
PERSPECTIVES

For QC - 36 Multi Contrast Targets with single injection - 6 times fewer experiments under same conditions



Customizable Solution: Bridges the gap to clinically realistic ground truths

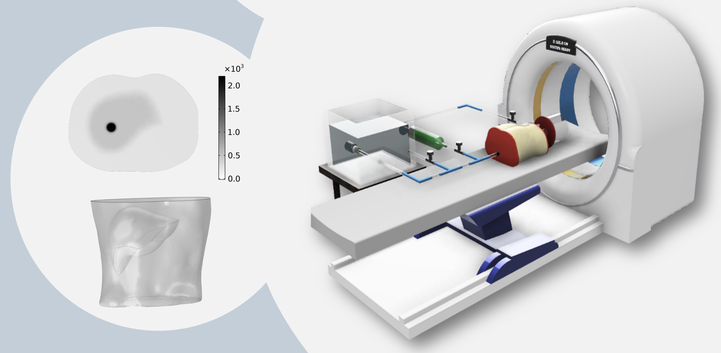
Validated Reliability: <10% signal deviation from physical ground truth



•Dynamic Imaging Extension: Time-activity variation for 4D imaging

Future Work

•Expanded Lesion Database: Integrating complex, **heterogeneous morphologies** (necrotic cores, irregular boundaries)



•**Strength:** Framework functions at complex textures and clinically realistic shapes
•**Limitation:** 3D Printing is limited to contrast ratios of **~5:1** due to printing limits
•**Solution: Simulation** component can model high-contrast target (>5:1)

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References